INFO 4220 – Networks II: Market Design
Spring 2022 – Problem Set 3

SOLUTION

Problem 1 (20 points)

Abby, Ben, Chloe have recently traveled to Fruit Island in search of quality fruits (Kiwis, Guavas, and Mangos). Naturally, each of these travelers have their own preferences over which of these fruits they desire most:

Abby	***Ben***	***Chloe***
Kiwi	*Kiwi*	*Mango*
Mango	*Guava*	*Guava*
Guava	*Mango*	*Kiwi*

a) Upon arrival, they are greeted by the Fruit King, who tries to offer a Kiwi to Abby, Guava to Ben, and Mango to Chloe (the Fruit King only has one fruit of each). Does this assignment scheme produce a Pareto efficient matching? Is this assignment scheme fair? (hint: think in the case in which Abby, Ben, and Chloe go every weekend to Fruit Island, and the Fruit King always allocate the fruits in the same way. Will anyone eventually object)

> *ANSWER: The assignment is efficient. The only person not getting his favorite fruit is Ben, and to make him better off we would need to take the Kiwi from Abby, which would make her worse off. The assignment is not fair (or using the same scheme always would not be fair) because Ben doesn't get a chance to get his favorite fruit.*

b) Noting that both Abby and Ben like Kiwis the most, The Fruit King instead decides to assign fruits to travelers via a random serial dictatorship. Find the random assignment of fruits to travelers via this scheme.

> *ANSWER: We need to consider all the possible orders and the assignments that would result from each of them.*

Orders			*Assignments*		
First	*Second*	*Third*	*Abby*	*Ben*	*Chloe*
Abby	*Ben*	*Chloe*	*Kiwi*	*Guava*	*Mango*
Abby	*Chloe*	*Ben*	*Kiwi*	*Guava*	*Mango*
Ben	*Abby*	*Chloe*	*Mango*	*Kiwi*	*Guava*
Ben	*Chloe*	*Abby*	*Guava*	*Kiwi*	*Mango*
Chloe	*Abby*	*Ben*	*Kiwi*	*Guava*	*Mango*
Chloe	*Ben*	*Abby*	*Guava*	*Kiwi*	*Mango*

The probabilistic assignment is:

	Abby	*Ben*	*Chloe*
Kiwi	*1/2*	*1/2*	*0*
Mango	*1/6*	*0*	*5/6*
Guava	*1/3*	*1/2*	*1/6*

c) Suppose instead that fruits are assigned via a probabilistic serial algorithm (eating algorithm). Find the resulting random assignment:

ANSWER: At time 0, Abby and Ben start eating the Kiwi while Chloe starts eating Mango. After some time, Abby and Ben finish the Kiwi. Each of them ate 1/2, and Chloe has eaten 1/2 of the Mango. Then, Abby starts eating Mango (along with Chloe), and Ben starts eating Guava. When Abby and Chloe finish the Mango, Abby ate 1/4, while Chloe ate 3/4. At this point Ben has eaten 1/4 of the Guava. Next, all continue eating Guava (the only fruit left). By the time they finish the Guava Abby ate 1/4, Ben 1/2, and Chloe 1/4. The random assignment then is:

	Abby	*Ben*	*Chloe*
Kiwi	*1/2*	*1/2*	*0*
Mango	*1/4*	*0*	*3/4*
Guava	*1/4*	*1/2*	*1/4*

d) Let's think about why one of these random matching schemes might be preferable to the other. Does either random serial dictatorship or probabilistic serial assignment stochastically dominate the other?

ANSWER: Let's check for each individual which of the assignments stochastically dominates.

P(Getting fruit as food as … or better)	*RSD*	*PS*
Abby		
Kiwi	*1/2*	*1/2*
Mango	*2/3*	*3/4*
Ben		
Kiwi	*1/2*	*1/2*
Mango	*1*	*1*
Chloe		
Mango	*5/6*	*3/4*
Guava	*1*	*1*

For Abby PS stochastically dominates RSD, Ben is indifferent between PS and RSD, and for Chloe RSD stochastically dominates PS. There is no clear dominance of one random assignment over the other.

e) Suppose that random serial dictatorship and probabilistic serial allocation are the only possible random assignments schemes that can be used. Are either of these two random assignment schemes ex-ante or ex-post efficient? Be sure to justify any statements you make. You may cite theorems used in class to justify your answers. When unable to do so, you should manually verify whether or not a given algorithm is ex-ante/post efficient.

ANSWER: We know from the Bogomolnaia and Moulin theorem that any assignment calculated using the probabilistic serial algorithm is ex-ante efficient. We also know that the random serial dictatorship will always produce ex-post efficient assignments (because we now serial dictatorships are efficient, and the random assignment produced using the random serial dictatorship are just combinations of assignment created using serial dictatorships).

We need check the assignment we found with the PS algorithm is efficient. We can decompose the random assignment in the following way:

$$\begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/4 & 0 & 3/4 \\ 1/4 & 1/2 & 1/4 \end{pmatrix} = \frac{1}{2} * \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} + \frac{1}{4} * \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \frac{1}{4} * \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

As all the deterministic assignments above are efficient, the assignment we found using the PS assignment is also ex-post efficient.

To know if the assignment we found with the random serial dictatorship is ex-ante efficient, recall the definition of ex-ante efficiency: "A random assignment β is ex-ante efficient (also called ordinally efficient) if there is no other random assignment β' such that, for each individual, β' stochastically dominates β." From the result of part d) and because the question tells you to assume that the two random assignments we found are the only two possible random assignments, you can conclude that the assignment you found using the random serial dictatorship is also ex-ante efficient.

Problem 2 (20 points)

The deferred acceptance algorithm is one of the methods that are usually employed to solve the school matching problem. However, there is one difficulty when using this algorithm: The algorithm requires schools' priorities over students to be strict, while in practice many schools don't have strict priorities over students. The solution to this problem is to employ a lottery to transform weak priorities (i.e. when students have the same priority at a given school) into strict priorities. In this problem we will explore different methods for implementing the lottery.

First, let's make some assumptions. There are 3 students and 3 schools. Each school has a capacity of 1 student. The preferences of students over schools are the following:

i_1	i_2	i_3
s_2	s_1	s_1
s_1	s_2	s_2
s_3	s_3	s_3

None of the individuals has any preferences at school 1. In school 2, i_1, i_2 have first priority, and i_3 has second priority. In school 3, i_3 has first priority, i_1 second, and i_2 last. We can summarize students' priorities as follows (individuals within [] have the same priority):

s_1	s_2	s_3
$[i_1, i_2, i_3]$	$[i_1, i_2]$	i_3
	i_3	i_1
		i_2

a) The first method we will try is to break the ties in the following way: Each school is going to (separately and independently) hold a lottery and determine the priority of students for that school. Find all possible strict priority orderings that may results with this method, and the assignments that would result from applying the deferred acceptance algorithm for the school problem (note: see appendix):

ANSWER: There are 12 possible orders that could happen.

Order 1.1			*Order 1.2*		
s_1	s_2	s_3	s_1	s_2	s_3
i_1	i_1	i_3	i_1	i_2	i_3
i_2	i_2	i_1	i_2	i_1	i_1
i_3	i_3	i_2	i_3	i_3	i_2
$\mu(i_1) = s_2, \mu(i_2) = s_1, \mu(i_3) = s_3$			$\mu(i_1) = s_2, \mu(i_2) = s_1, \mu(i_3) = s_3$		

Order 2.1			*Order 2.2*		
s_1	s_2	s_3	s_1	s_2	s_3
i_1	i_1	i_3	i_1	i_2	i_3
i_3	i_2	i_1	i_3	i_1	i_1
i_2	i_3	i_2	i_2	i_3	i_2
$\mu(i_1) = s_2, \mu(i_2) = s_3, \mu(i_3) = s_1$			$\mu(i_1) = s_1, \mu(i_2) = s_2, \mu(i_3) = s_3$		

Order 3.1			*Order 3.2*		
s_1	s_2	s_3	s_1	s_2	s_3
i_2	i_2	i_3	i_2	i_1	i_3
i_1	i_1	i_1	i_1	i_2	i_1
i_3	i_3	i_2	i_3	i_3	i_2
$\mu(i_1) = s_2, \mu(i_2) = s_1, \mu(i_3) = s_3$			$\mu(i_1) = s_2, \mu(i_2) = s_1, \mu(i_3) = s_3$		

Order 4.1			*Order 4.2*		
s_1	s_2	s_3	s_1	s_2	s_3
i_2	i_2	i_3	i_2	i_1	i_3
i_3	i_1	i_1	i_3	i_2	i_1
i_1	i_3	i_2	i_1	i_3	i_2
$\mu(i_1) = s_2, \mu(i_2) = s_1, \mu(i_3) = s_3$			$\mu(i_1) = s_2, \mu(i_2) = s_1, \mu(i_3) = s_3$		

Order 5.1			*Order 5.2*		
s_1	s_2	s_3	s_1	s_2	s_3
i_3	i_1	i_3	i_3	i_2	i_3
i_1	i_2	i_1	i_1	i_1	i_1
i_2	i_3	i_2	i_2	i_3	i_2
$\mu(i_1) = s_2, \mu(i_2) = s_3, \mu(i_3) = s_1$			$\mu(i_1) = s_3, \mu(i_2) = s_2, \mu(i_3) = s_1$		

Order 6.1			*Order 6.2*		
s_1	s_2	s_3	s_1	s_2	s_3
i_3	i_2	i_3	i_3	i_1	i_3
i_2	i_1	i_1	i_2	i_2	i_1
i_1	i_3	i_2	i_1	i_3	i_2
$\mu(i_1) = s_3, \mu(i_2) = s_2, \mu(i_3) = s_1$			$\mu(i_1) = s_2, \mu(i_2) = s_3, \mu(i_3) = s_1$		

b) A simpler way to solve this problem is to do just 1 lottery for all schools. The ordering determined by this lottery is then used to break ties in all schools (for example, if the central lottery resulted in the order i_3, i_2, i_1, the priorities in school 1 will be i_3, i_2, i_1 and at school 2 will be i_2, i_1, i_3). Find all the strict priority orderings that may result with this method, and the assignments that would result from applying the deferred acceptance algorithm for the school problem (note: you can just re-use some parts of part a):

ANSWER: This simply corresponds to the assignments we found with orders x.1, with $x \in [1,6]$ *in part (a)*

c) Let's now try to determine if one method is better than the other. Look at the possible assignments that result from part a) and part b). Are there assignments that can only happen with one of the methods? If there are any, analyze if those assignments are stable with respect to the original priorities (i.e. when checking if there is justifiable envy you need to look at the original weak priorities, not the strict priorities you generated by breaking ties)? Are they efficient?

ANSWER: Let's make a table of the different assignments that happen, and how many times they happen when we use multiple tie breaking (each school run their lottery), or single tie breaking (only one lottery is run for all schools)

Assignment	*# Times in MTP*	*# Times in STP*	i_1	i_2	i_3
μ	*6*	*3*	s_2	s_1	s_3
μ'	*3*	*2*	s_2	s_3	s_1
μ''	*1*	*0*	s_1	s_2	s_3
μ'''	*2*	*1*	s_3	s_2	s_1

Note that all assignments are individually rational and non-wasteful, so when checking for stability we only need to determine if there is justifiable envy.

Let's see if μ is efficient and stable:

- *It is efficient, because the only individual not matched to their preferred option is i_3 and the only way to make them better is making i_1 or i_2 worse*
- *It is stable, because while i_3 would prefer s_1 or s_2, at s_1 all students have the same priority, and in s_2, i_3 has the lowest priority*

⇒ μ is efficient and stable

Let's see if μ' is efficient and stable:

- *It is efficient, because the only individual not matched to their preferred option is i_2 and the only way to make them better is making i_1 or i_3 worse*
- *It is stable, because while i_2 would prefer s_1 or s_2, at s_1 they all have the same priority, and in s_2, i_1 (who is assigned to s_2) and i_2 have the same priority. Thus there is no justified envy*

⇒ μ' is efficient and stable

Let's see if μ'' is efficient and stable:

- *It is **NOT EFFICIENT**, because if i_1 and i_2 switched assignments they would both be better off*
- *It is stable, because while i_2 would prefer s_1, at s_1 they all have the same priority, and while i_1 would prefer s_2, at s_2, i_2 (who is assigned to s_2) and i_1 have the same priority. Thus there is no justified envy*

⇒ μ'' is not efficient, but it is stable

Let's see if μ''' is efficient and stable:

- *It is efficient, because: i_2 would like to have s_1 instead, but that would require taking s_1 away from i_3 and that would make i_3 worse off; i_1 would prefer s_1 or s_2, but giving i_1 wither of those options would make i_2 or i_3 worse off*
- *It is stable, because while i_2 would prefer s_1, and at s_1 they all have the same priority. i_1 would prefer s_1 or s_2, but at s_1 all student have the same priority, and in s_2, i_2 (who is assigned to s_2) and i_1 have the same priority. Thus there is no justified envy*

⇒ μ''' is efficient and stable

Note that the only assignment that happens with MTP and that doesn't happen in STP is μ'' and that isn't an efficient assignment.

d) Now let's think in terms of probabilistic assignments. Express the assignments you found in part a) and part b) in terms of random assignments and determine if either stochastically dominates the other

ANSWER: With MTP (Separate lotteries) the random assignment β_{MTB} is (note rows are individuals and columns are schools):

$$\begin{pmatrix} 1/12 & 3/4 & 1/6 \\ 1/2 & 1/4 & 1/4 \\ 5/12 & 0 & 7/12 \end{pmatrix}$$

With STP (Only one lottery) the random assignment β_{STB} is

$$\begin{pmatrix} 0 & 5/6 & 1/6 \\ 1/2 & 1/6 & 2/6 \\ 1/2 & 0 & 1/2 \end{pmatrix}$$

For i_1 β_{STB} stochastically dominates β_{MTB} as it gives a slightly better chance of getting the first choice (5/6 vs 3/4), and the same chance of getting the first or second choice (10/12).

For i_2 β_{MTB} stochastically dominates β_{STB} as it gives the same chance of getting the first choice (1/2), and a slightly better chance of getting the first or the second choice (3/4 vs 2/3)

It is easy to see that i_3 prefers β_{STB} to β_{MTB}

e) The school board is asking your advice regarding which method to use. What would you recommend and why?

ANSWER: In this case, there is no reason to prefer having multiple lotteries, because when we use MTP we sometimes get assignments that are not efficient, and in terms of stochastic dominance there isn't one assignment that dominates the other, but for two individuals STP dominates MTP, and for only 1 individual MTP dominates STP.

Problem 3 (1 point each, no partial credit)

Please answer each of these questions. Your answers should be short and precise (as a rule of thumb, no answer should take you more than 2 or 3 lines to explain).

3.1. A principal is designing a market mechanism to allocate three different radio spectrum bands between three different agents. The agents have different preferences over which radio spectrum band they like. To allocate the bands the principal uses the following procedure:

a) Asks each agent for a number between 1 and 10
b) The principal draws a random number between 1 and 10
c) The principal lets the agent that chose the number closest to the random draw choose the radio spectrum band they like better (if there is a tie, the tie is resolved by playing rock, paper, scissors)

d) The principal lets the agent that chose the second closest number to the random draw choose the radio spectrum band they like better among those remaining (if there is a tie, the tie is resolved by playing rock, paper, scissors)
e) The principal lets the agent that chose the third closest number to the random draw choose the radio spectrum band they like better among those remaining (if there is a tie, the tie is resolved by playing rock, paper, scissors)

Is this an example of a direct or an indirect mechanism? Explain why.

ANSWER: This is an example of a direct mechanism because the agents reveal their private information. The number asked by the principal in the beginning is only to determine the order they pick. When they pick, they have to reveal the band they prefer.

3.2 Is the mechanism outlined in 3.1 strategy proof? Explain why.

ANSWER: Yes, it is strategy proof because when it is the turn of an agent to choose, they only action that makes sense is for them to reveal their true preference (if they lie and say their preference is something different, they get a worse band)

3.3 In the re-design of the Boston Public School assignment mechanism they considered using the deferred acceptance algorithm or the top trading cycle algorithm. Explain in which case you would prefer one or the other (assume schools have priority orderings over students)

ANSWER: In the school matching problem there is a conflict between stability and efficiency. If you want to achieve efficiency, you should use the top trading cycle algorithm. If you want to achieve stability you should use the deferred acceptance algorithm.

3.4 The Boston Public School system was using the immediate acceptance algorithm to assign students to schools. Why was this mechanism bad, and why did some people still defend its use?

ANSWER: The immediate acceptance algorithm is not strategy proof. Parents have incentives to strategize and list higher schools they think they have a good chance of accepting their kids. Some people like it because it gives the appearance of matching students to their top choices (the problem is the preference submitted are not really their top choices)

3.5 Describe the rural hospital problem. Can this problem be solved (i.e. can you find a mechanism that produce an stable matchings that resolves the rural hospital problem)?

ANSWER: The rural hospital problem is when some hospitals don't fill up their vacancies in an assignment problem. It cannot be solved if using an stable mechanism. The rural hospital theorem says that any hospital that doesn't fill its vacancies in an stable matching, will not fill its vacancies in any stable matching.

3.6 If you have an assignment problem with public endowments (and without priorities over objects). Which algorithm is better: The serial dictatorship algorithm or the top trading cycle algorithm? Explain why

ANSWER: When there are no endowments you can use either algorithm and will obtain efficient assignments. To use the top trading cycle algorithm in a problem without endowments you just assign the objects to anyone initially and then run the TTC (in a similar way in which in the serial dictatorship you just create an order of individuals)

3.7 Consider the college dorm room assignment problem. If the housing department is using a serial random dictatorship with squatting rights, should returning students (i.e. students that already have a room assigned) participate in the market or keep their room?

ANSWER: They shouldn't always participate, because a random serial dictatorship with squatting rights does not guarantee that the assignment is going to be individually rational. A students that choses to participate may end up with a room they like less than their previous room.

3.8 In a medical match, what is the problem of using a mechanism that doesn't produce stable results?

ANSWER: When a mechanism that doesn't produce stable matching is used, the market tends to unravel as doctors and hospitals start preferring to match outside of the market

3.9 What is the difference between a matching problem and an assignment problem? What implications does this difference has for evaluating the efficiency of matchings/assignments?

ANSWER: In a matching problem you are matching agents to agents. In an assignment problem you match agents to objects. When evaluating efficiency in a matching, you have to consider the preference of both sides. When you evaluate the efficiency of an assignment, you only care about preferences of agents (as objects don't have preferences)

3.10 Explain what does it mean that an assignment in an assignment problem with private endowments is in the core? What should individuals do if an assignment is not in the core?

ANSWER: It means that you cannot find a coalition of agents that would be better off by not participating in the market and trading between themselves. If an assignment is not in the core, some individuals should choose not to participate and trade their initial endowments between them instead.